

Don't Red Team AI like a Chump

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@adversariel

Intro

- Who am I?
- What is this talk?

A story about hacking an AI system IRL



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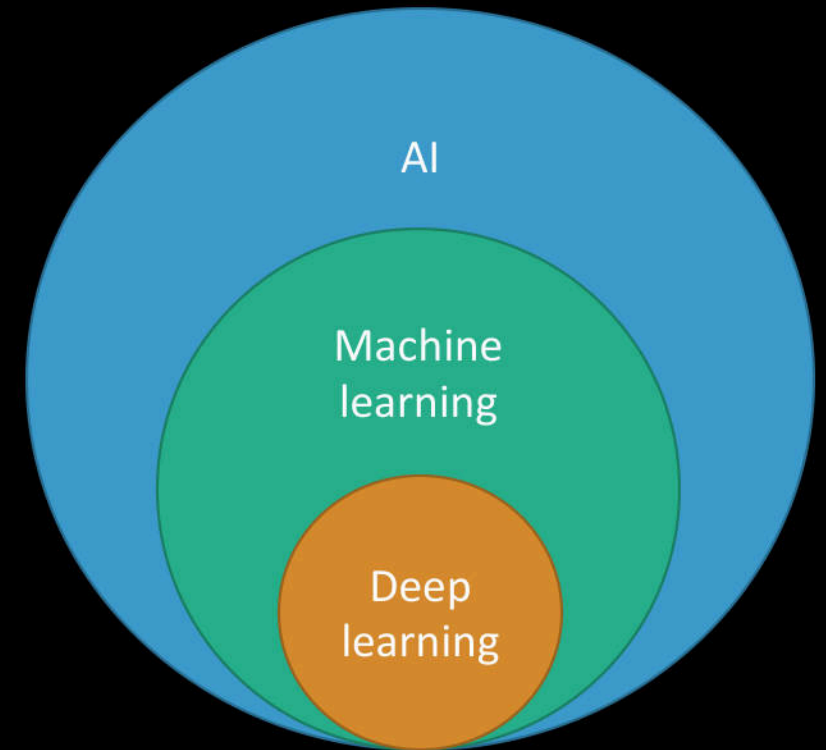


Key points in this talk (in case you zone out)

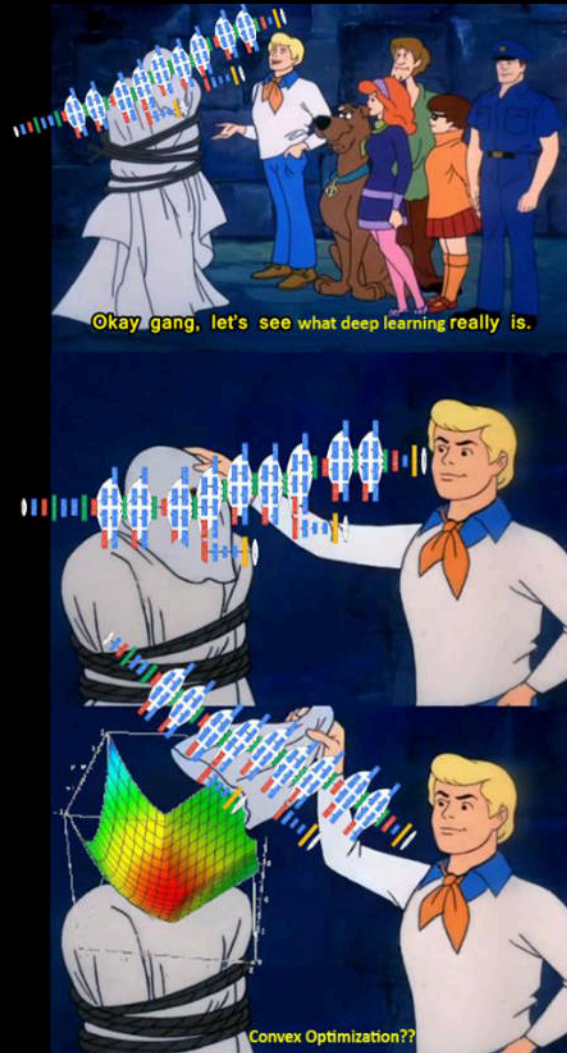
- **Focus on the threat model**
- Go for the ~~jugular~~ feature extraction process
- Think about the data supply chain

What is an AI system?

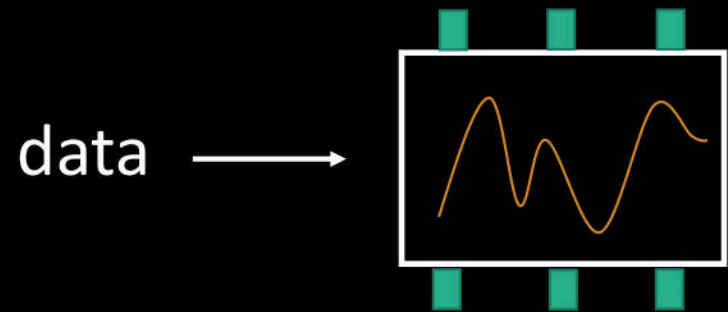
- AI is a class of algorithms that we use to extract actionable information from data
- AI is not new and the hype is real
- In this talk, AI == ML



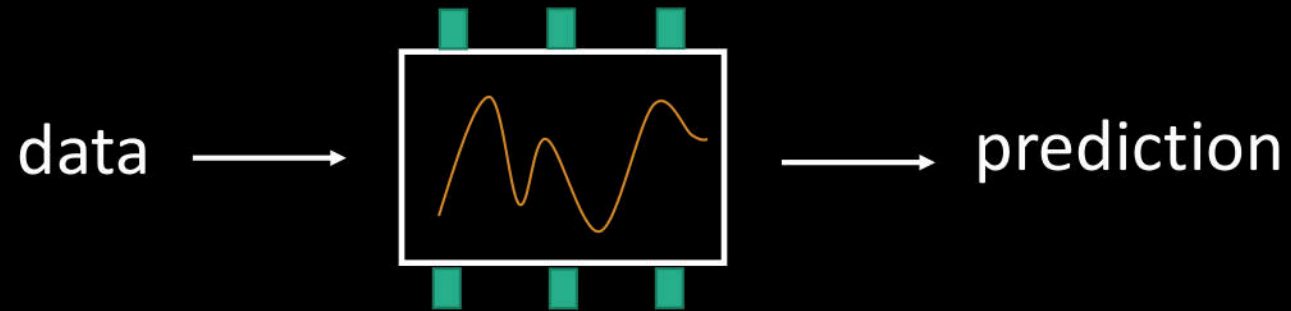
What is an AI system?



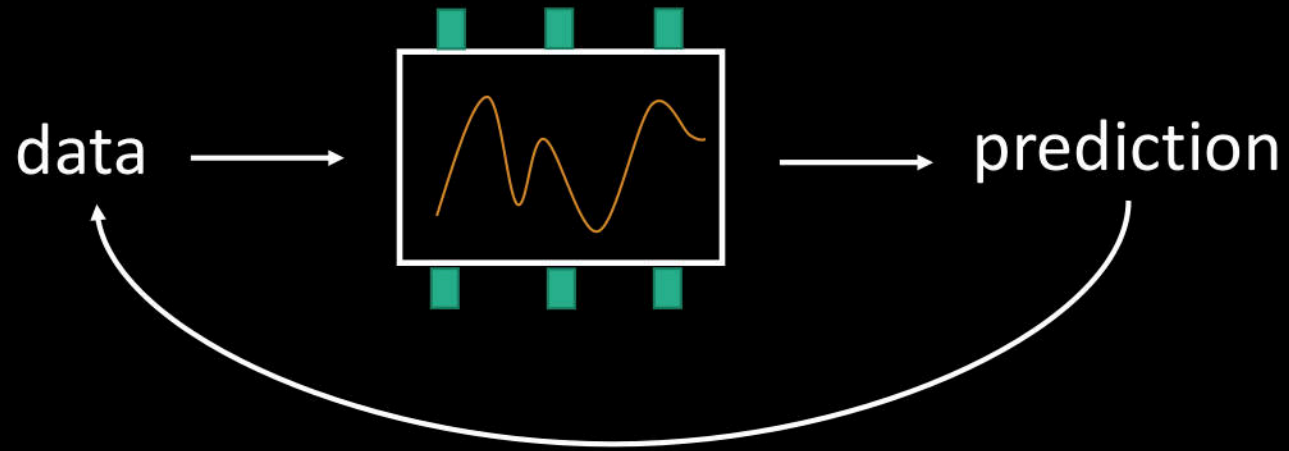
What is an AI system?



What is an AI system?



What is an AI system?



Parts of an AI system we can exploit

- Data
 - Training/testing sets
 - Deployed environment data
- Model
 - Algorithm
 - Parameters

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- data poisoning
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Parts of an AI system we can exploit

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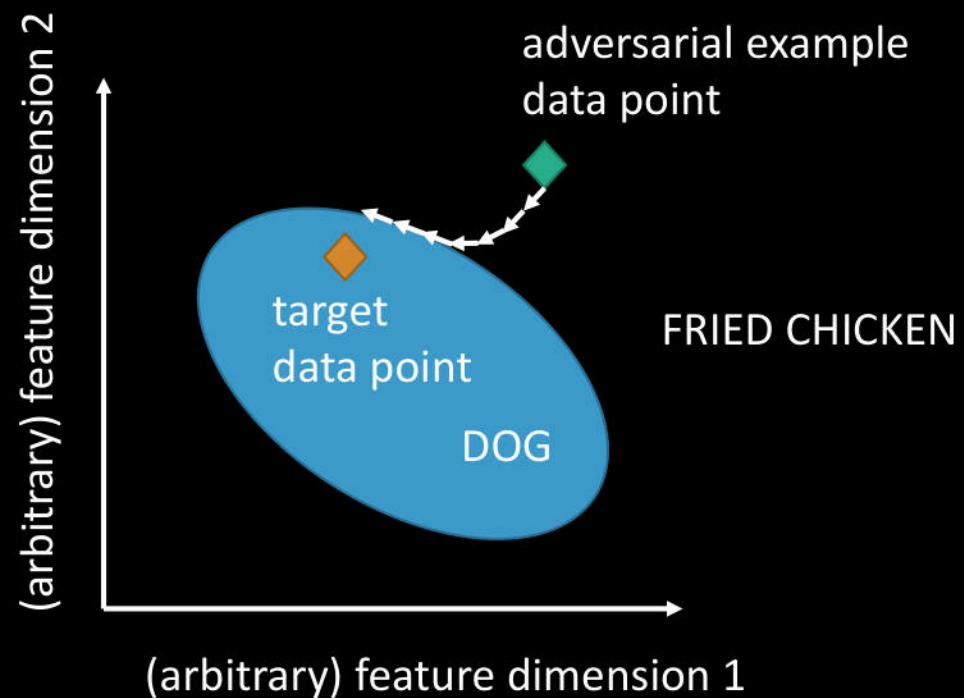
data poisoning



- Model

- Algorithm
- Parameters

Adversarial examples



Parts of an AI system we can exploit

- Data
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- backdoored function
- 

Parts of an AI system we can exploit

- Data

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model inversion



```
graph LR; MI(model inversion) --> TTS(Training/testing sets); MI --> P(Parameters);
```

The diagram illustrates the concept of model inversion. Two orange arrows originate from the text 'model inversion' on the right. One arrow points to 'Training/testing sets' under the 'Data' category, and the other points to 'Parameters' under the 'Model' category.

Model inversion



Image of Bill:
exists in training data set

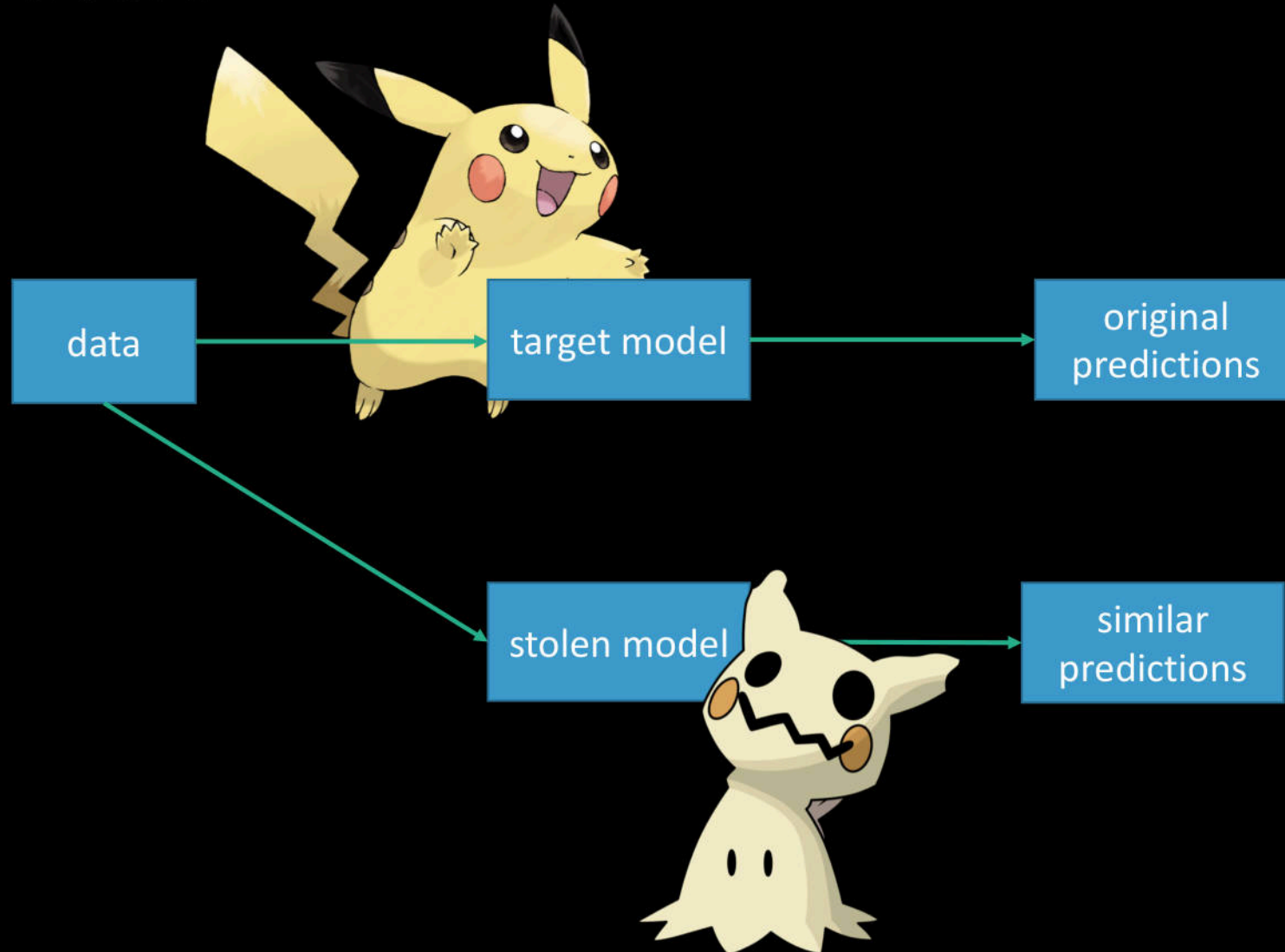


Recovered image

Parts of an AI system we can exploit

- Data
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- ← model theft

Model theft

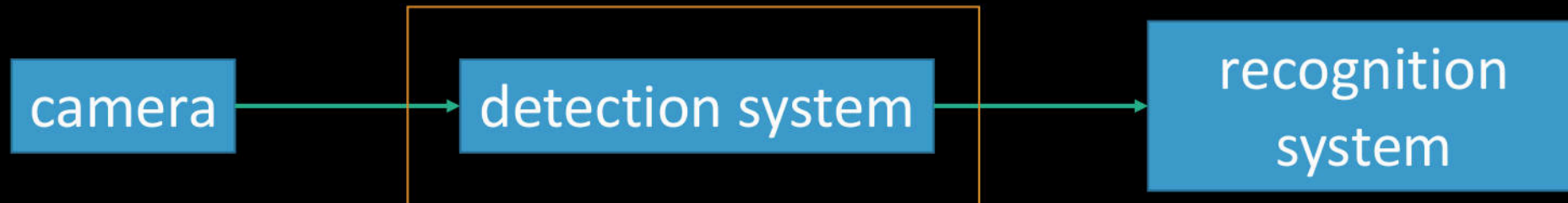


How to design an attack for an AI system

1. What model are you attacking?
2. Where do the data come from?
3. Where do the predictions go?

Fooling AI-powered video surveillance

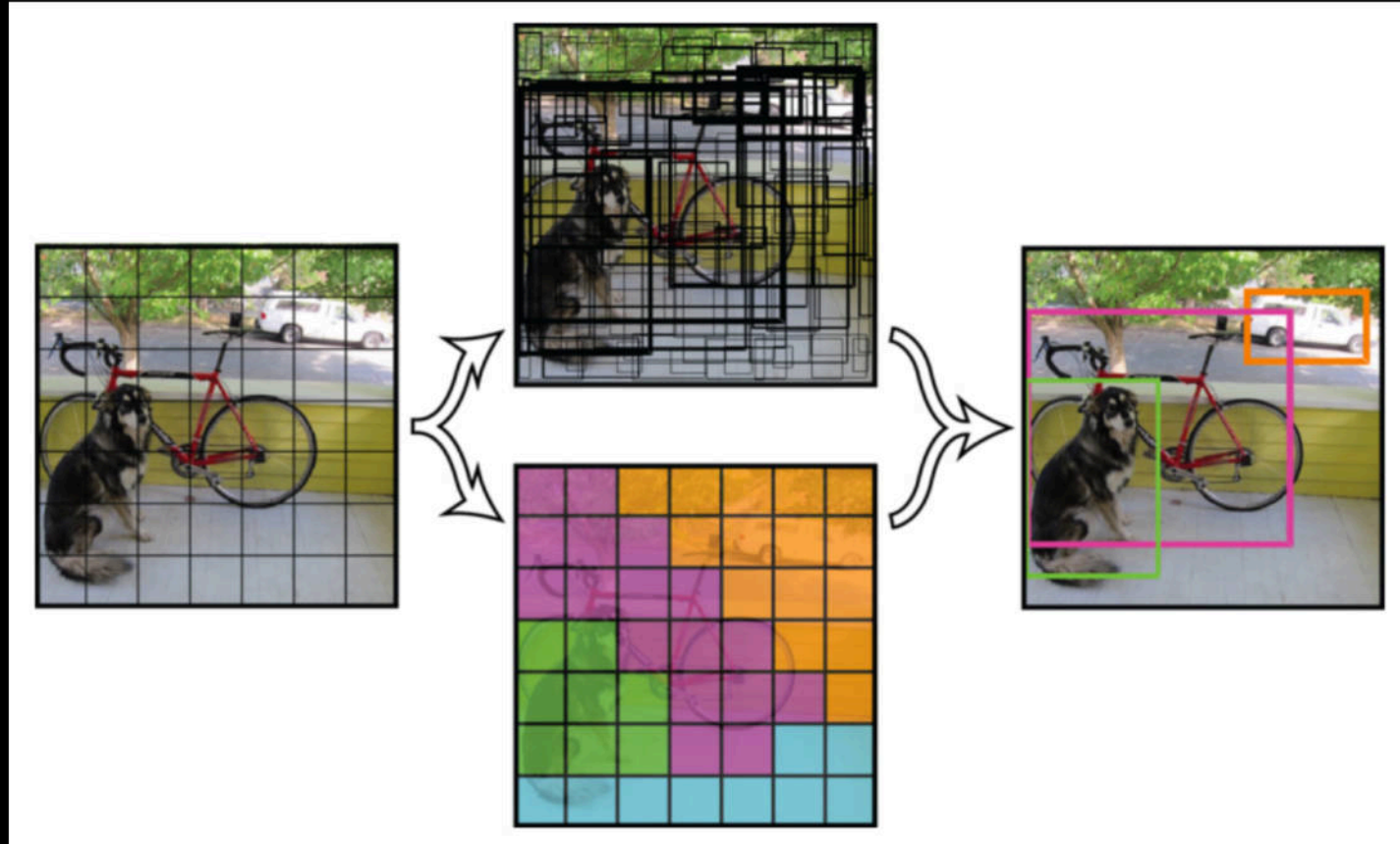
- Common system components
 - Detection: on-premises algorithm that checks each still image to see if there is a face for it to forward to the recognition system
 - Recognition: off-site look-up of flagged images using a data base of known faces



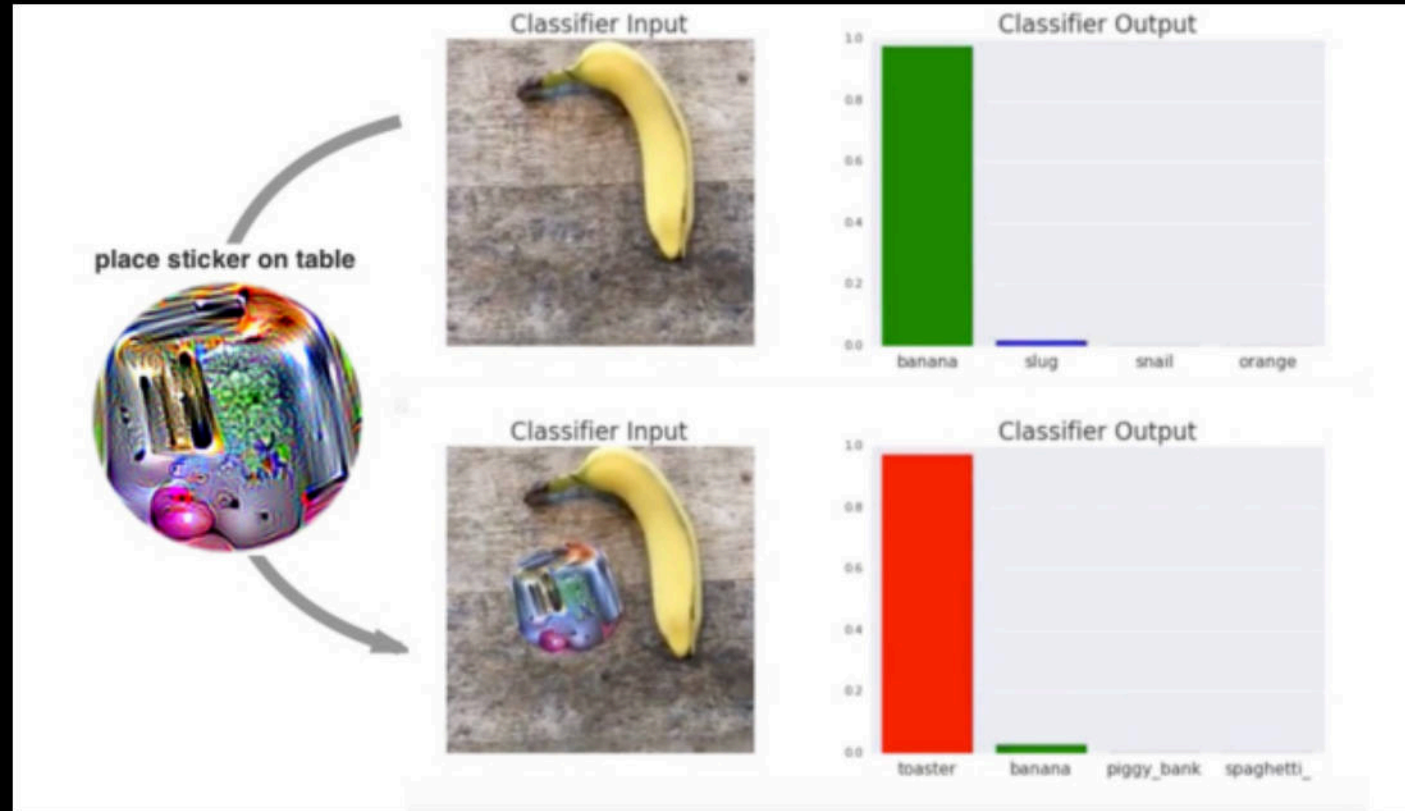
Fooling AI-powered video surveillance

1. What model are you attacking? YOLO v2
2. Where do the data come from? video frames
3. Where do the predictions go? stored as set of flagged frames

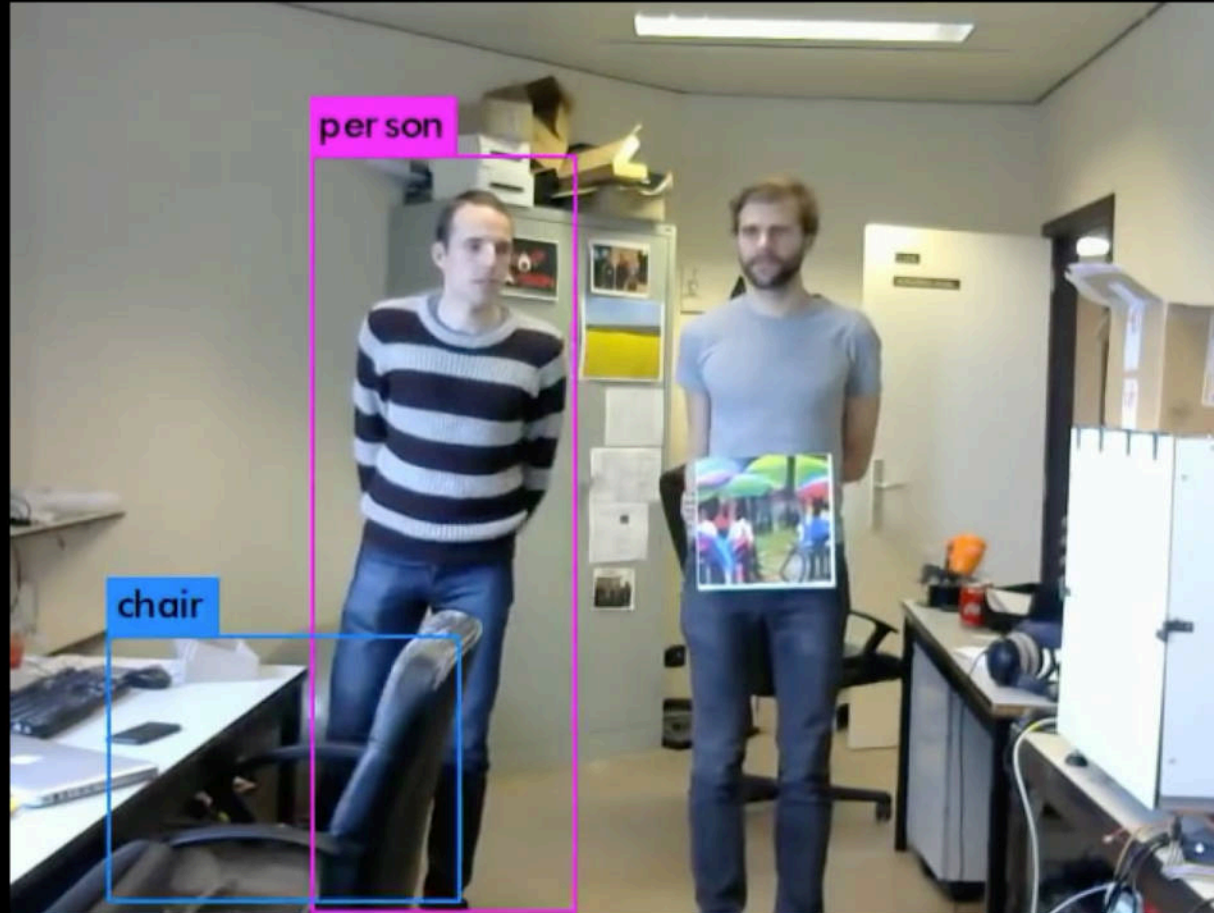
Fooling AI-powered video surveillance



Fooling AI-powered video surveillance



Fooling AI-powered video surveillance



AI red teaming is out of whack

- Red teaming AI is often conflated with the academic discipline of adversarial machine learning

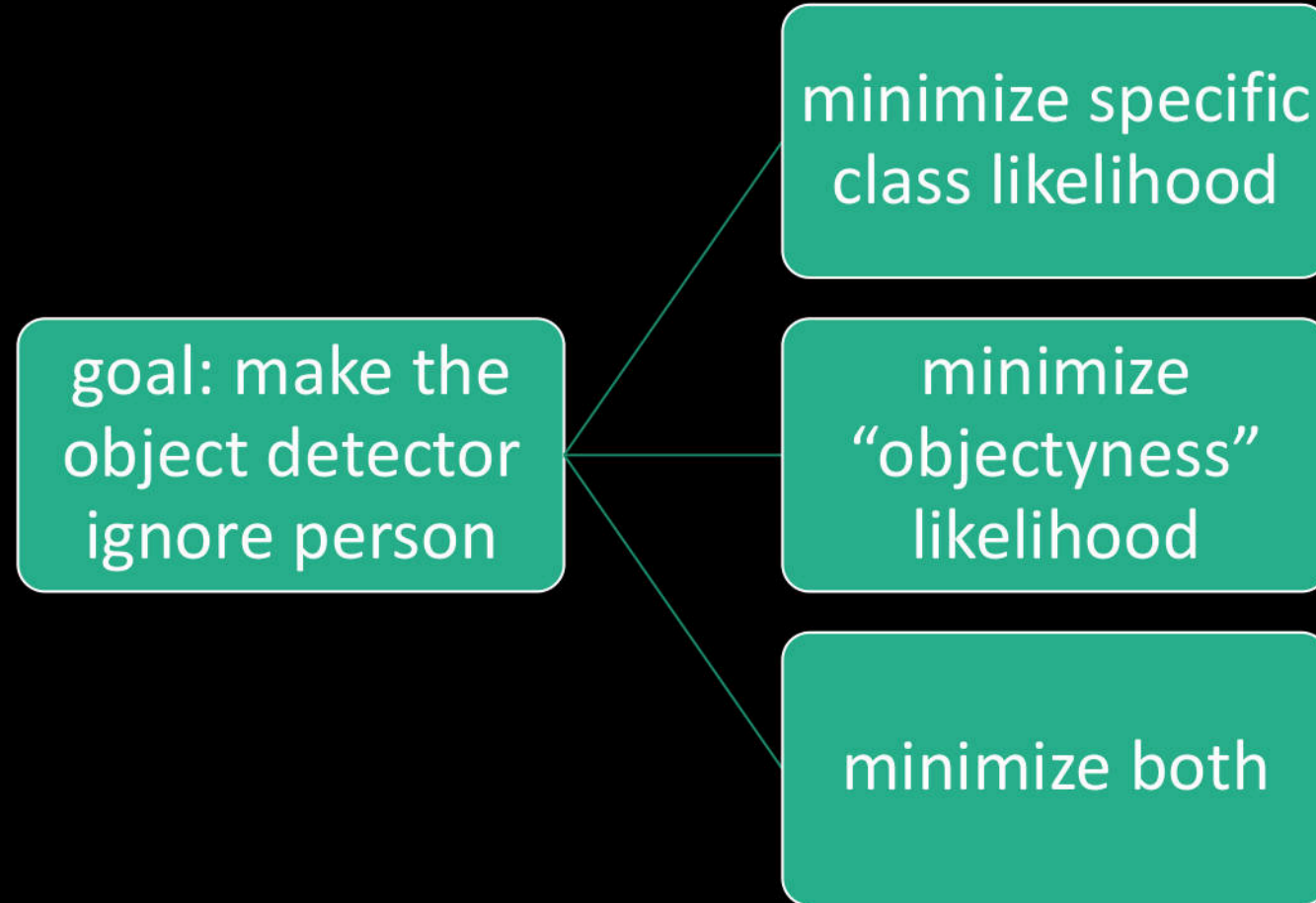
Cool ways to attack an AI model with math
== adversarial ML

Evaluate the security of an AI system
== red teaming AI



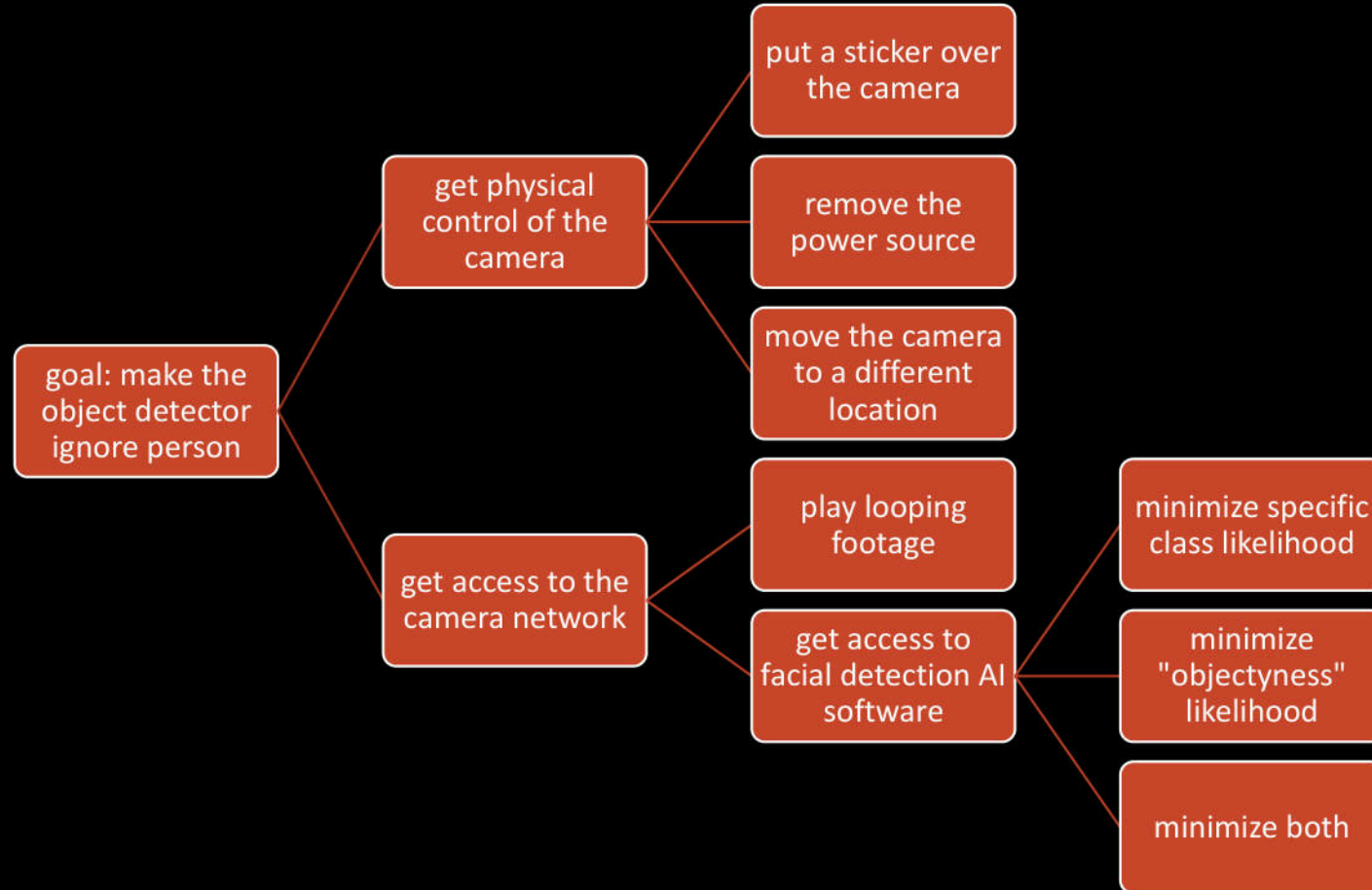
Adversarial ML attack tree

Fooling AI-powered video surveillance



AI red team attack tree

Fooling AI-powered video surveillance



AI red team attack tree

Fooling AI-powered video surveillance



Adversarial ML attack tree

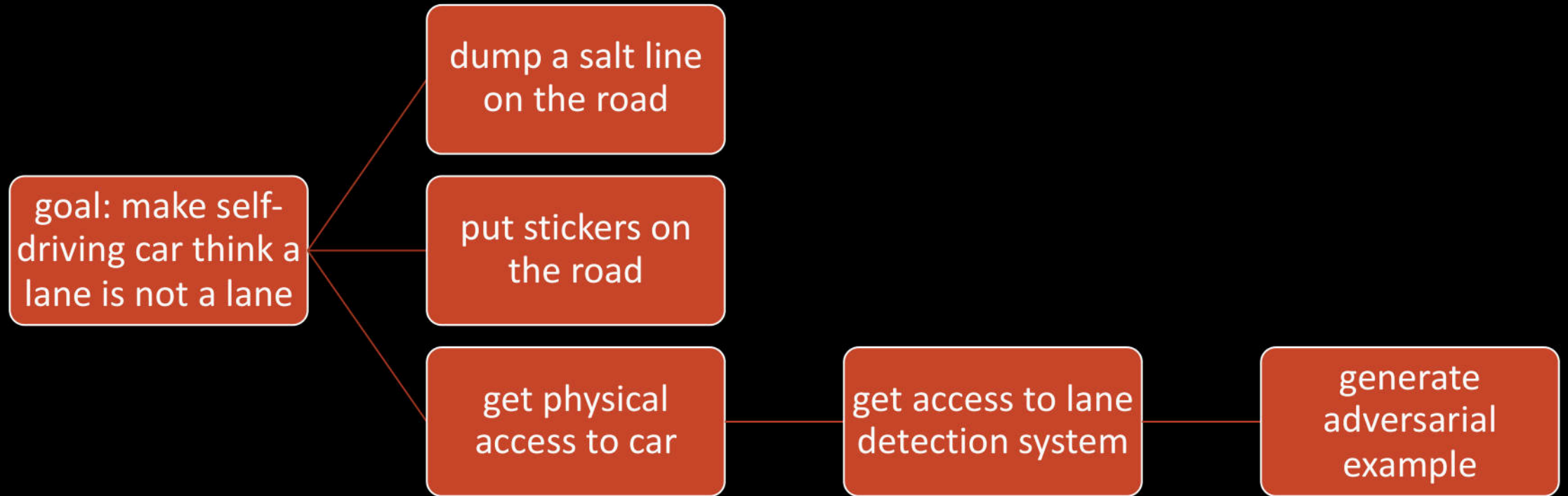
Breaking self-driving car lane detection

goal: make self-driving car think a lane is not a lane

generate adversarial example

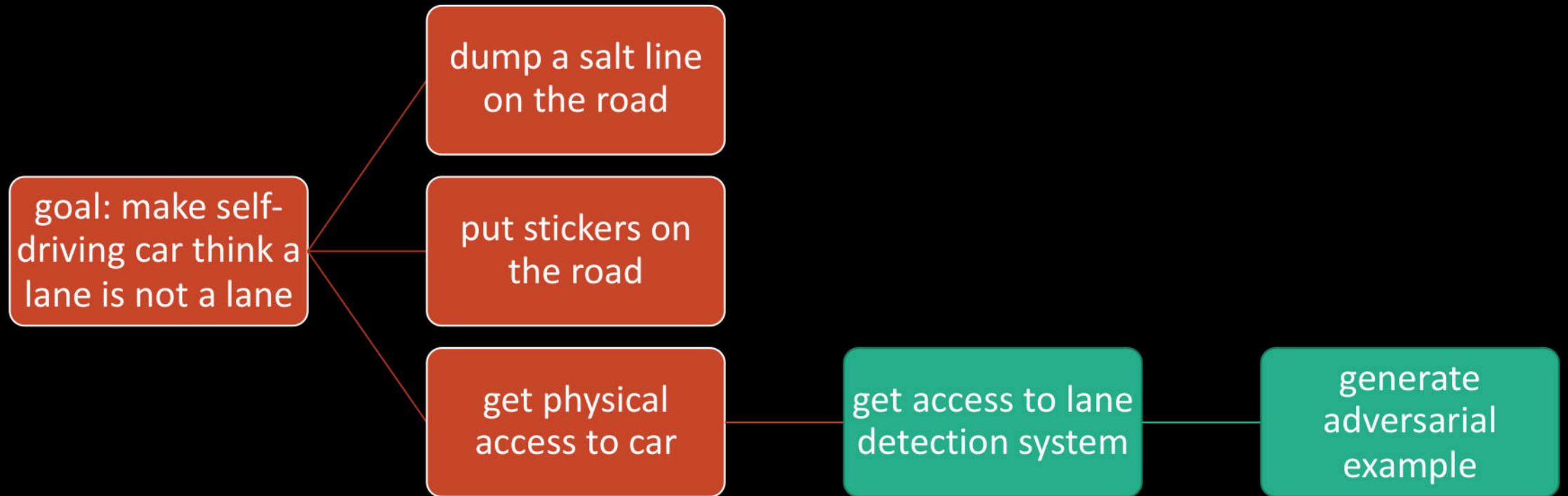
AI red team attack tree

Breaking self-driving car lane detection



Adversarial ML attack tree

Breaking self-driving car lane detection



How to design an attack for an AI system

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How to design an attack for an AI system

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How to design an attack for an AI system

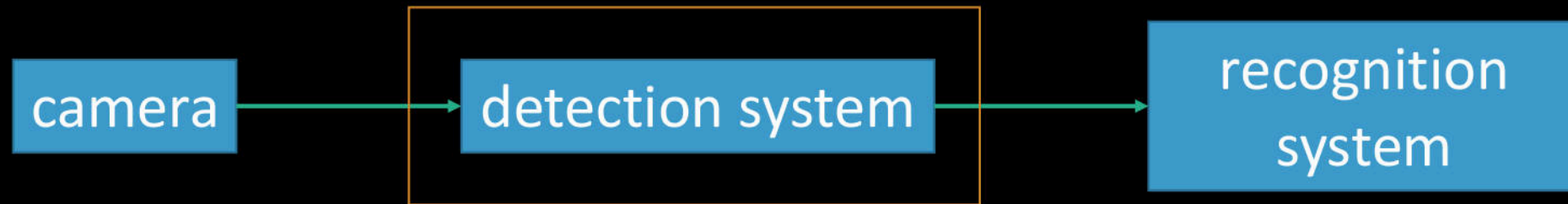
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2. What is the data processing pipeline?
 - i. Where do the data come from?
 - ii. What is the data representation?
 - iii. What is the output?

How to design an attack for an AI system

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How to design an attack for an AI system

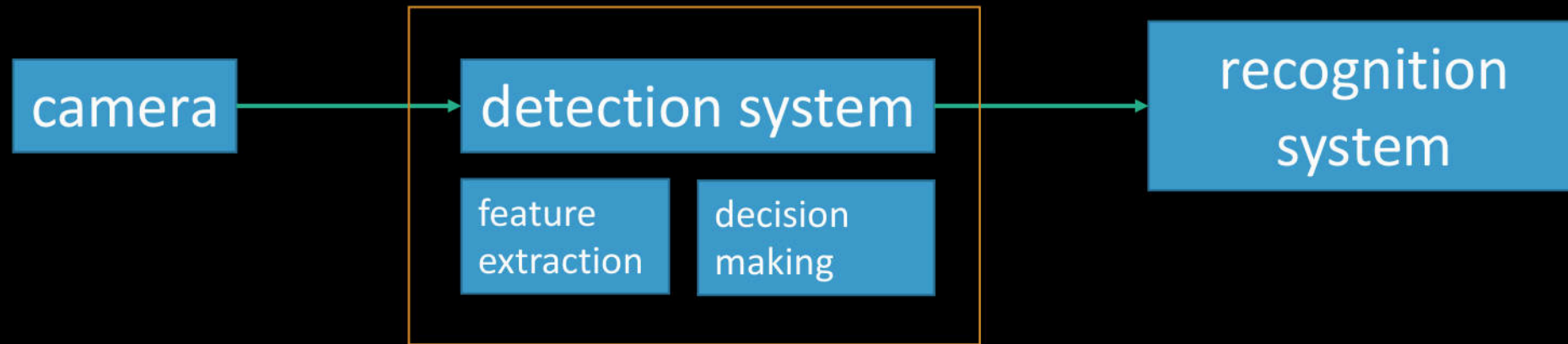
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How to design an attack for an AI system

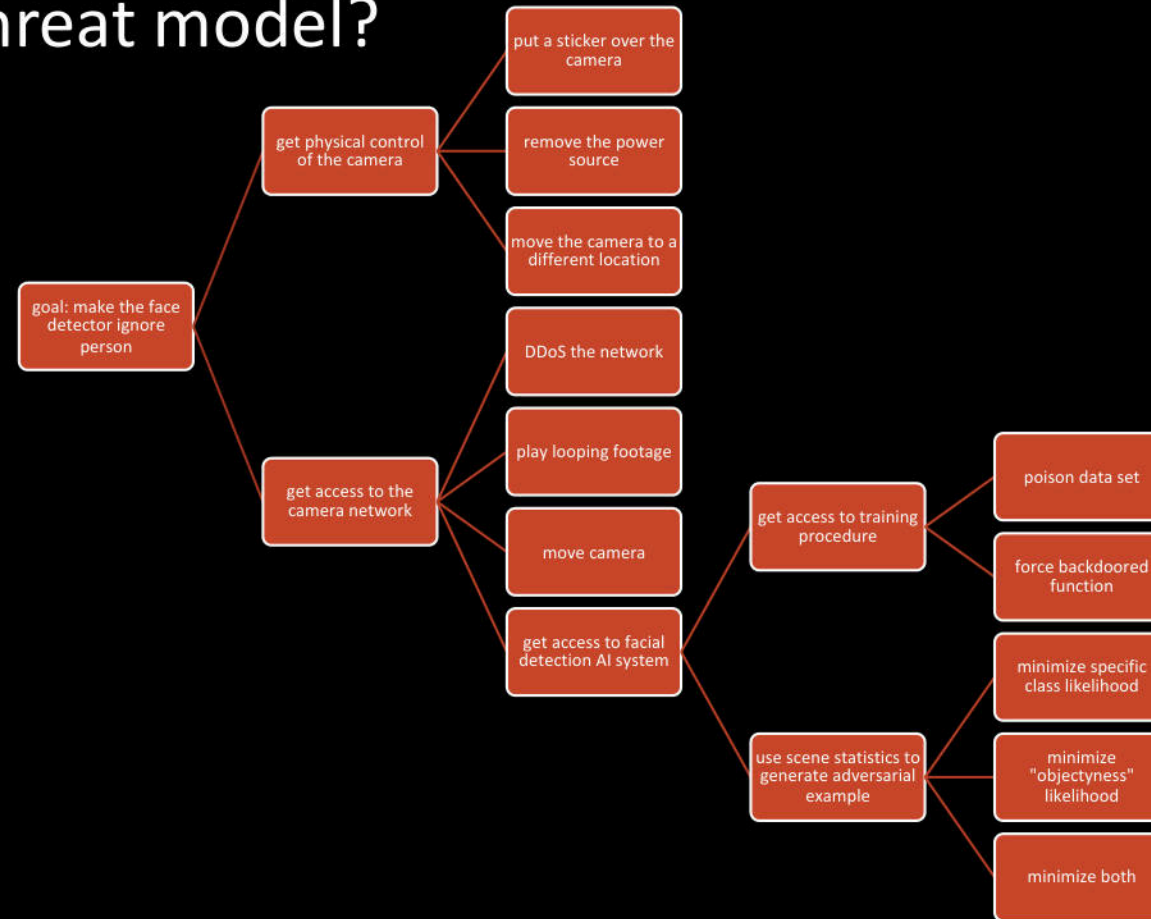
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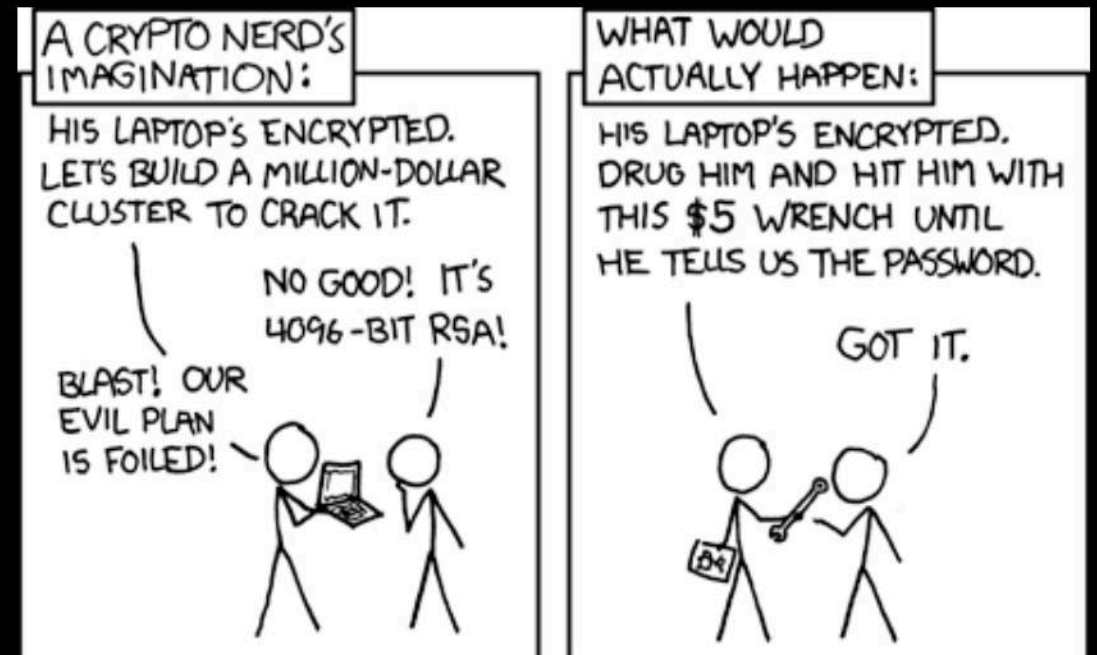
How to design an attack for an AI system

3. What is the threat model?



Tl;dw

- Go for the ~~jugular~~ feature extraction process
- Think about the data supply chain
- Focus on the threat model



For more information (and math!)

Feel free to reach out via Twitter (@adversariel) and/or grab a beer with me :)

- Attacks:

- Tencent Keen Security Lab, 2019 - https://keenlab.tencent.com/en/whitepapers/Experimental_Security_Research_of_Tesla_Autopilot.pdf
- [Biggio et al, 2018] - <https://pralab.diee.unica.it/sites/default/files/biggio18-pr.pdf>
- [Fredrikson et al, 2015] - <https://www.cs.cmu.edu/~mfredrik/papers/fjr2015ccs.pdf>
- [Tramèr et al, 2016] - <https://arxiv.org/pdf/1609.02943.pdf>
- [Thys et al, 2019] - <https://arxiv.org/pdf/1904.08653.pdf>
- [Brown et al, 2017] - <https://arxiv.org/pdf/1712.09665.pdf>

- Threat modeling:

- Schneier, 1999 - https://www.schneier.com/academic/archives/1999/12/attack_trees.html